

- Precision reference resistor

To verify that the design is within the ADC range of operation, Calculate the voltages for AIN1 and AIN2 and the maximum differential input voltage. Verify that V_{AIN1} and V_{AIN2} are within the input range of the PGA given the gain setting and supply voltage. Use the maximum RTD resistance based on the desired temperature measurement.

$$V_{AIN1} = I_{IDAC1} \cdot (R_{RTD} + R_{REF}) \quad (65)$$

$$V_{AIN2} = I_{IDAC1} \cdot R_{REF} \quad (66)$$

Additionally, verify that the voltage seen at the IDAC pin (where $V_{AIN0} = V_{AIN1}$) is within the current source compliance voltage. When the IDAC output voltage rises too close to AVDD, the IDAC loses compliance and the excitation current is reduced.

The reference resistor, R_{REF} must be a precision resistor with high accuracy and low drift. Any error in the R_{REF} reflects the same error in the RTD measurement. The REFP0 and REFN0 pins are shown connecting to the R_{REF} resistor as a Kelvin connection to get the best measurement of the reference voltage. This eliminates any series resistance as an error from the reference resistance measurement.

2.6.4 Measurement Conversion

$$\text{Output code} = 2^{23} \cdot \text{Gain} \cdot V_{RTD} / V_{REF} = 2^{23} \cdot \text{Gain} \cdot (I_{IDAC1} \cdot R_{RTD}) / (I_{IDAC1} \cdot R_{REF}) = 2^{23} \cdot \text{Gain} \cdot R_{RTD} / R_{REF} \quad (67)$$

$$R_{RTD} = R_{REF} \cdot \text{Output code} / (2^{23} \cdot \text{Gain}) \quad (68)$$

2.6.5 Configuration Register Settings

For the measurement, set the register values:

- Select multiplexer settings for AINP and AINN to measure leads 2 and 3 of the RTD
- Enable the PGA, set gain to desired value
- Select data rate and digital filter settings
- Select reference input to measure R_{REF} for ratiometric measurement
- Enable the internal reference (the IDAC requires an enabled internal reference)
- Set IDAC magnitude and select IDAC1 output pin to drive lead 1 of the RTD

2.7 Two Series Two-Wire RTD Measurements, Low-Side Reference

A single IDAC may drive multiple RTDs at the same time. In this schematic, two two-wire RTDs are stacked on top of each other for measurement through different inputs of the ADC multiplexer.